

Problem Set 07 Discussion

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Kayla Wilhoit

Housekeeping

- Reminder to include the Data Lunch reflection writeup in your problem set submission!
 - 5% of grade, will be entered at end of semester
 - Can upload separate reflection document later to your repository instead of editing the notebook if needed
- Final project proposals are due **Nov 4th** (two weeks from today)
 - Please reach out early if you are having trouble finding examples!
- Feel free to email me (kayla.wilhoit@duke.edu) for any questions about problem sets, the final project, or using class concepts in your research!
 - Our lab has a code starter kit (work in progress) with guides for setting up VS Code, Globus for file transfers, and ssh keys for connecting to the DCC

<https://github.com/magwenelab/CodeStarterKit>

Data Wrangling

- **select()** statements vs. **rename()**

```
names(data)[1] <- "disease"
```

```
rename(disease="...1")
```

```
select("disease" = "...1", "case_fatality_rate" = "case fatality rate",
```

- **gsub()** and **rename_with()**

- Remove % sign with mutate:

```
mutate(case_fatality_rate = gsub("%", "", case_fatality_rate))
```

- Replace all spaces with underscores in colnames:

```
rename_with(~ gsub(" ", "_", .x))
```

- **ggrepel** will automatically drop labels if a region is much too overplotted – can set a higher threshold manually or resize the plot

```
Warning: ggrepel: 9 unlabeled data points (too many overlaps). Consider  
increasing max.overlaps
```